

MAHAD JAFRI

AI Engineer | LLM Agents | Production ML Systems

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PROFESSIONAL SUMMARY

AI Engineer with 5+ years building and deploying production LLM agents, RAG systems, and ML pipelines in Python. Specialize in agentic architectures (LangGraph, ReAct, HITL), retrieval-augmented generation, and the data infrastructure to run them reliably at scale.

Proven impact: **\$20M+** in client revenue lifted, **500K+** daily inferences served, **90%** reduction in manual ops.

CAREER HIGHLIGHTS

- **AI & Automation:** Shipped a driver-incentive optimization agent (LangGraph + RL) that auto-generates targeted activation campaigns, cutting campaign costs by **60%** and ops time by **90%** vs. manual workflows.
- **Business Growth:** Contributed to **\$20M+** in client revenue across telco and mobility, including **\$10M** lifted on Telefónica Spain through Bayesian pairing models serving 500K real-time inferences/day.
- **Top Performer:** Recognized in the **top 10%** of Afiniti employees for **4 consecutive years** for technical leadership and revenue impact.

EXPERIENCE

Analytics Engineer, Sadapay (Remote)

Mar 2026 – Present

- Owning **end-to-end data operations**, financial reporting automation, and self-serve tooling for operational and executive teams.

Senior Analytics & Automation Lead, Careem (Uber) (Dubai/Remote)

Mar 2025 – Mar 2026

Real-Time Supply-Balancing Agent (Forecasting + RL)

- Built an autonomous **LangGraph** agent that forecasts driver supply gaps with **LSTM and gradient-boosted models** and proposes targeted incentive campaigns via **reinforcement learning**, cutting campaign costs by **60%** at **WMAPE <10%**.
- Shipped the agent into ops workflows via a **Slack-integrated Human-in-the-Loop (HITL) approval flow** with checkpointing and state recovery, reducing manual execution time by **90%**.

Text-to-SQL Agent & Schema-Aware SQL Copilot

- Built a **ReAct-based Text-to-SQL agent** with schema retrieval, query validation, and self-correction loops; deployed via internal API to reduce executive ad-hoc data requests by **40%+**.
- Developed a **VS Code extension** for schema-aware SQL completion using LLM tool calling and embedding-based table retrieval, accelerating analyst pipeline development by **50%**.

Senior Data Scientist, Afiniti Software Solutions, (USA/Remote)

Mar 2020 – Mar 2025

- **Production ML at Scale:** Led a team of 4 to deliver **\$10M annual revenue lift** for Telefónica Spain; trained pairing-optimization models serving **~500K real-time inferences/day** on Afiniti's 24/7 production platform.
- **Revenue Forecasting:** Built a 90-day forecasting pipeline using gradient-boosted ensembles with **85% accuracy** from 3-day behavioral signals to flag early-stage churn and automate LTV reporting.
- **Bayesian Modeling:** Productionized MCMC (Stan) and decision-tree pairing models that lifted client revenue by **5%**, validated through controlled A/B experiments.
- **Infrastructure & Tooling:** Enhanced training and validation modules for Afiniti's company-wide R modeling platform, **improving global model validation accuracy by 10%**.
- **Data Engineering:** Optimized multi-stage PostgreSQL/MySQL attribution logic, **improving outcome-to-call match rate by 8%** and **reducing processing latency by 40%**.

TECHNICAL SKILLS

LLMs & GenAI: LLM Agents, LangGraph, LangChain, ReAct, RAG, Tool Use, Prompt Engineering, Text-to-SQL, Embeddings, Vector DBs (pgvector, FAISS), Fine-tuning, Hugging Face, OpenAI/Anthropic APIs

Machine Learning: Deep Learning, Reinforcement Learning, LSTMs, Gradient Boosting (XGBoost, LightGBM), Random Forest, MCMC (Stan), Time-Series Forecasting, A/B Testing, NLP

Engineering & MLOps: Python, SQL (PostgreSQL, MySQL, Redshift), R, C++, PyTorch, scikit-learn, pandas, FastAPI, Docker, Git, CI/CD, ETL Orchestration, Superset, Tableau, Power BI

EDUCATION

Bachelor of Science, Electrical Engineering

Lahore University of Management Sciences, 2019